

Outlines



- **General causes of Abnormal Karyotypes**
- **Sex Determination**
- **Fertility as affected by chromosome**
- **Genetic material**

General causes of Abnormal Karyotypes:



- 1- The genetic make-up of the individual can be a predisposing factor
e.g. Bloom's syndrome in man (autosomal recessive disorder)
- 2- Ionizing radiation
- 3- Different types of drugs and chemicals as:
 - Alkylating agents
 - Radical producing agents
 - Basic dyes
 - Antimetabolites of DNA and its precursors
 - Few antibiotics such as (azaserine, mitomycin C and streptonigrin)
- 4- Certain viruses and mycoplasma are also known to cause structural damage to chromosomes

Sex Determination

Dr. Walaa Slouma
Professor of Genetics



26/10/2024

Sex Determination



In livestock, sex is determined by the sex chromosomes

► In all species except poultry → Male XY → heterogametic

Female XX → homogametic

► In poultry



Male ZZ → homogametic

Female ZW → heterogametic

Y chromosome is necessary to determine maleness in mammals

as in human, XO is known as

Turner's syndrome

→ Female, but they are infertile and have infantile reproductive systems

- XO in horses causes similar characteristics

Sex Determination



- While **XO** individuals in *Drosophila* (Fruit fly)

↓
are sterile males



↓
In insects, the Y chromosome is not necessary for **maleness** but is necessary for **fertility**

- In some species of insect as grasshopper

↓
Normally Male XO and Female XX



Mechanism of Sex Determination



XX-XO



Grasshopper

XX-XY



In all mammals, some plant, insects, reptiles

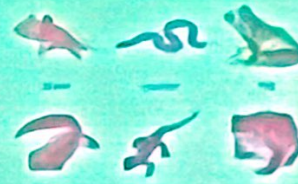
ZZ-ZW



Birds



Moths, some amphibians and Fishes



Sex Determination



- ▶ Since sex is determined basically by → a balance between the X and Y chromosomes in mammals

So, deviation than this usually result in sterility and/or other phenotypic modification

Ex.

Klinefelter's syndrome in man → 47,XXY

- Male internal and external genitalia
- May or may not have breast development
- Have small testes
- Sterile

Similar in cattle →

- Small testes
- Sterility

Sex Determination



- **The Y chromosome and male development in mammals**
- **X Chromosome Inactivation**
- **Foetal Sex Diagnosis**

The Y chromosome and male development in mammals

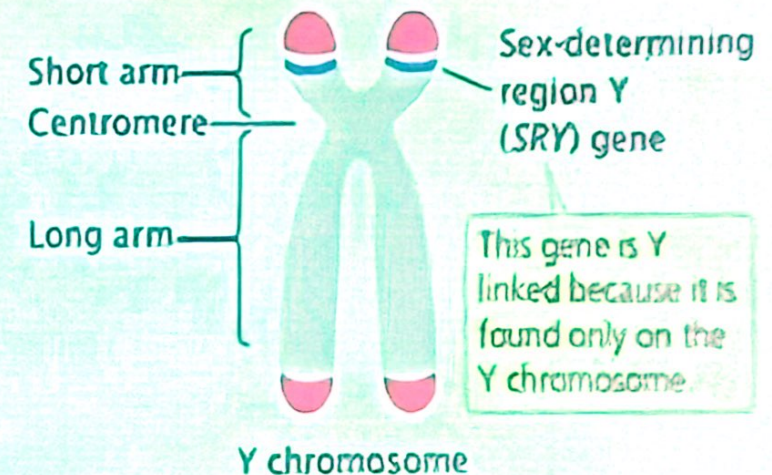


Y chromosome is necessary to determine **maleness** in mammals

Contain **genetic information** that responsible for **Testis determining factor (TDF)**, also known as **sex-determining region Y (SRY) gene**

That is trigger the undifferentiated gonadal tissue of embryo to form **testes**

This is responsible for causing **maleness**



X Chromosome Inactivation



❖ In most mammals

Female sex chromosomes



two large X chromosomes (XX)

Male sex chromosomes

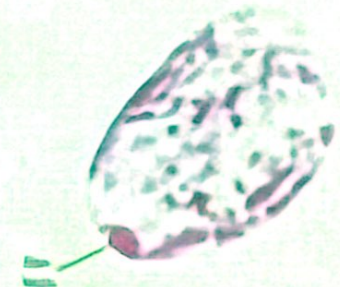
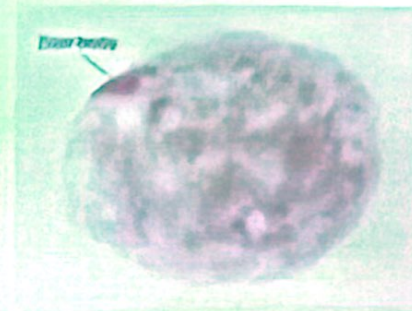


a large X chromosome and a small Y chromosome (XY)

- When there is more than one X chromosome, only one X chromosome and the other one is inactive but not totally

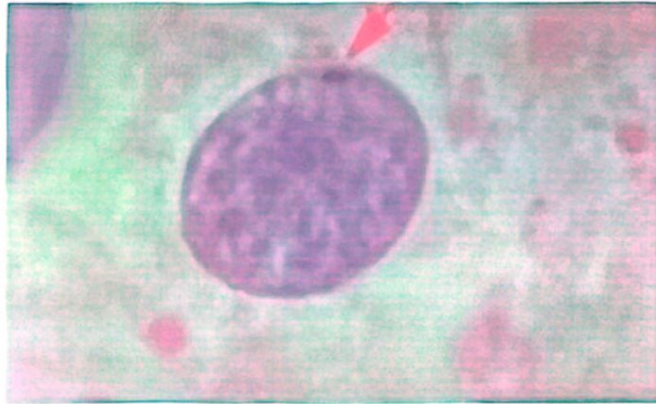


Called barr body or sex chromatin

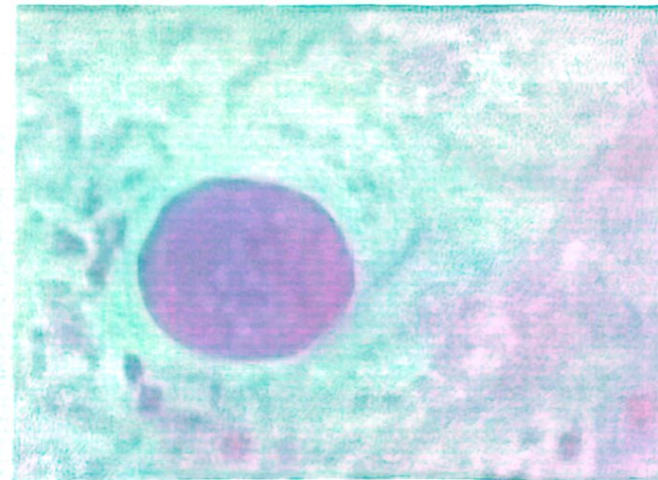


Prof. Dr. Walter S. S. S.

X Chromosome Inactivation



Female XX



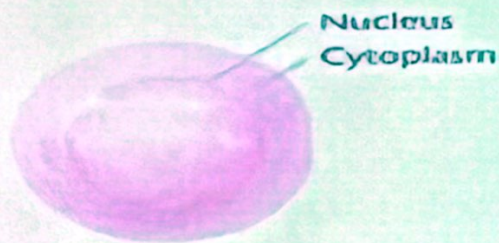
Male XY

X Chromosome Inactivation

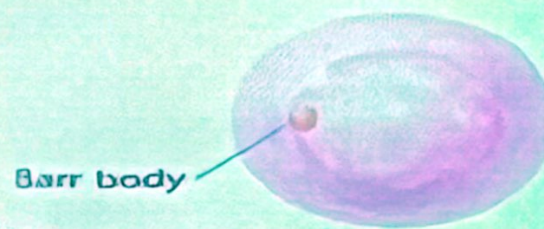


❖ In case of absence of one X ($2n-1,X$) → sexual differentiation and body growth are affected

❖ In case of duplicate in the male ($2n+1,XXY$) → it affects normal testicular development



46, X Y
45, X



46,  X
47,  XY



47,   X
48,   XY

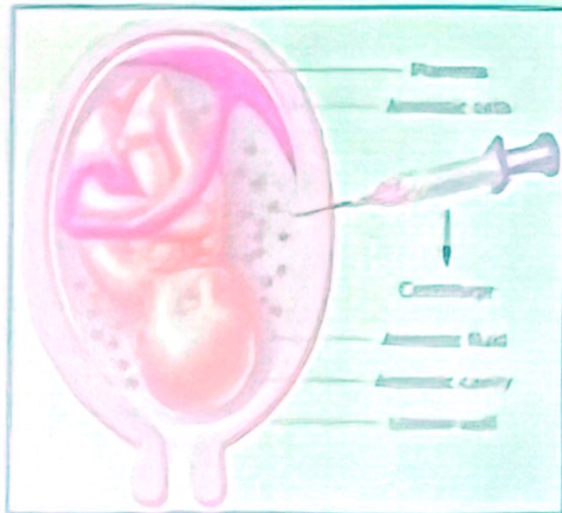
Foetal Sex Diagnosis



- Prenatal cytogenetic studies have been done on amniotic fluid cell cultures obtained by transabdominal **amniocentesis**

Amniocentesis

→ is a medical procedure used in prenatal diagnosis of chromosomal abnormalities, fetal infections, and sex determination



→ The fetal DNA is examined for genetic abnormalities

Foetal Sex Diagnosis



In high-risk groups as:

- Mothers over 35 years of age for whom the chance of having a child with one of the trisomy syndrome is increased
- Mothers who are carriers of X-linked recessive disorders



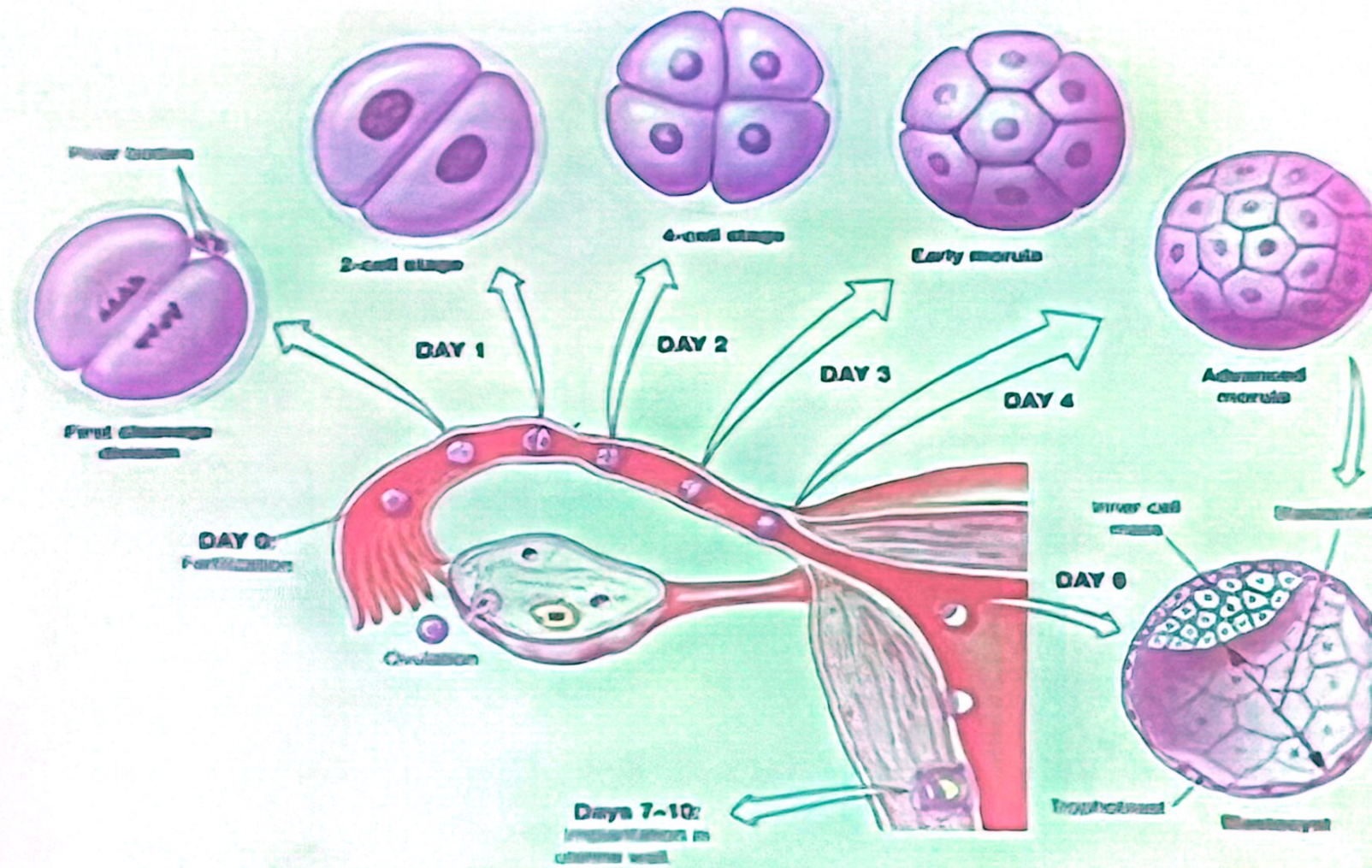
desirable therapeutically to abort male fetuses

In veterinary medicine

- Amniocentesis for cytogenetic purposes is practical only in large animals
But carried out in a bitch at 40 days of pregnancy
- In cattle, for example, fetal sex determination before 120 days gestation

Optimum time → 70 and 90 → fetal fluid 400 ml → 1000 ml

So, 10-15 ml of fetal fluids can safely be withdrawn without fluid replacement



Another way of sex determination

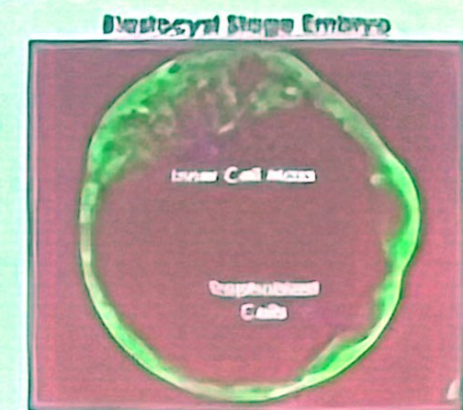
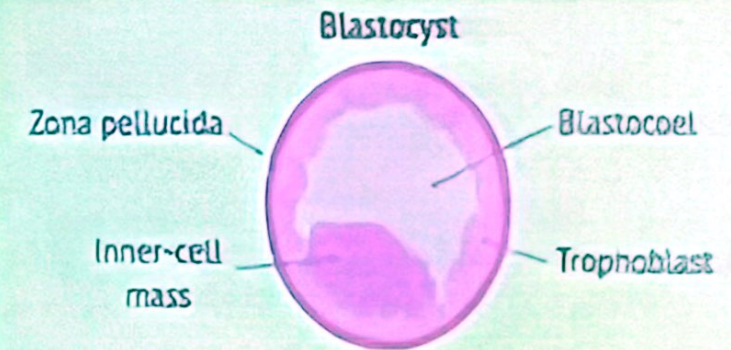


- In bovine embryo → Chromosomal analysis at time of embryo transfer from donor to recipient



Small fragment of trophoblast without damage the inner cell mass of blastocyst

- ✓ Sexing embryos during transfer has been done in sheep
- ✓ It also holds possibilities for other species, but the task is more difficult because the sex chromosomes are not as readily distinguished from the autosomes as they are in cattle



Prof. Dr. Veneta Stoyanova

Fertility as affected by chromosome



Genes or chromosomal aberrations can affect fertility by:

Direct effect

One of any hereditary factors which influence fertility

Indirect effect

Modifying the capability of the animal to react to variable management or environmental factors

Cases



1. Freemartinism

2. Intersex

1. Freemartinism



➤ Recognized in **cattle**



As a type of infertility associated with anatomical defects of reproductive tract in a heifer born twin to a bull

➤ **Cause** → Choriovascular anastomosis of placenta



Mixture of **XX** and **XY** cells in the developing fetus

Hormonal effect of the earlier developing male reproductive organs

Female (90% born with male)

Gonads

- Develop into **apparently normal ovaries** that are capable of ovulation
- Develop into **small testes**
- One or both gonads are classified as **ovotestes** in most cases

Reproductive tract

- External genitalia → are usually normal **except** for the **clitoris** which often **enlarged**
- Internally →
 - **regression** of Mullerian duct derivatives
 - **over-development** of Wolffian duct derivatives

Male

- **little** or **no** effect on the structure of male gonads and reproductive tract
- There is an effect on $\longrightarrow$ Reproductive ability
 $\downarrow$ due to
Decreased **sperm motility** and **sperm concentration**

➤ **In both male and female, diagnoses by**



Blood culture to check for the presence of XX/XY leukocyte chimaerism

Freemartins in Other Species (occur occasionally)

- Sheep, goats, horses and pigs → **but** incidence is much lower than in cattle

↓
exchange of cells
occurs in only 1%
of twins

↘
exchange of cells is common,
but not leads to infertility

- In birds

- Vascular anastomosis is common within double-yolked eggs
- Major end result is a feminization of the male reproductive tract

2. Intersex (Hermaphrodite)



➤ Intersex individuals, having a graded mixture of maleness and femaleness

➤ Intersex can also be classified into:

Chromosomal Intersex

Abnormal sexual development is due to:



abnormalities in sex chromosomes



Ex. - Klinefelter syndrome (xxy)
- Turner syndrome (xo)

Gonadal Intersex

Individuals having a normal sex chromosome But not crossponding gonad

Ex.

XX → with testes only

X Y → with ovaries only

XX, X Y → with ovatestes
(Freemartins)

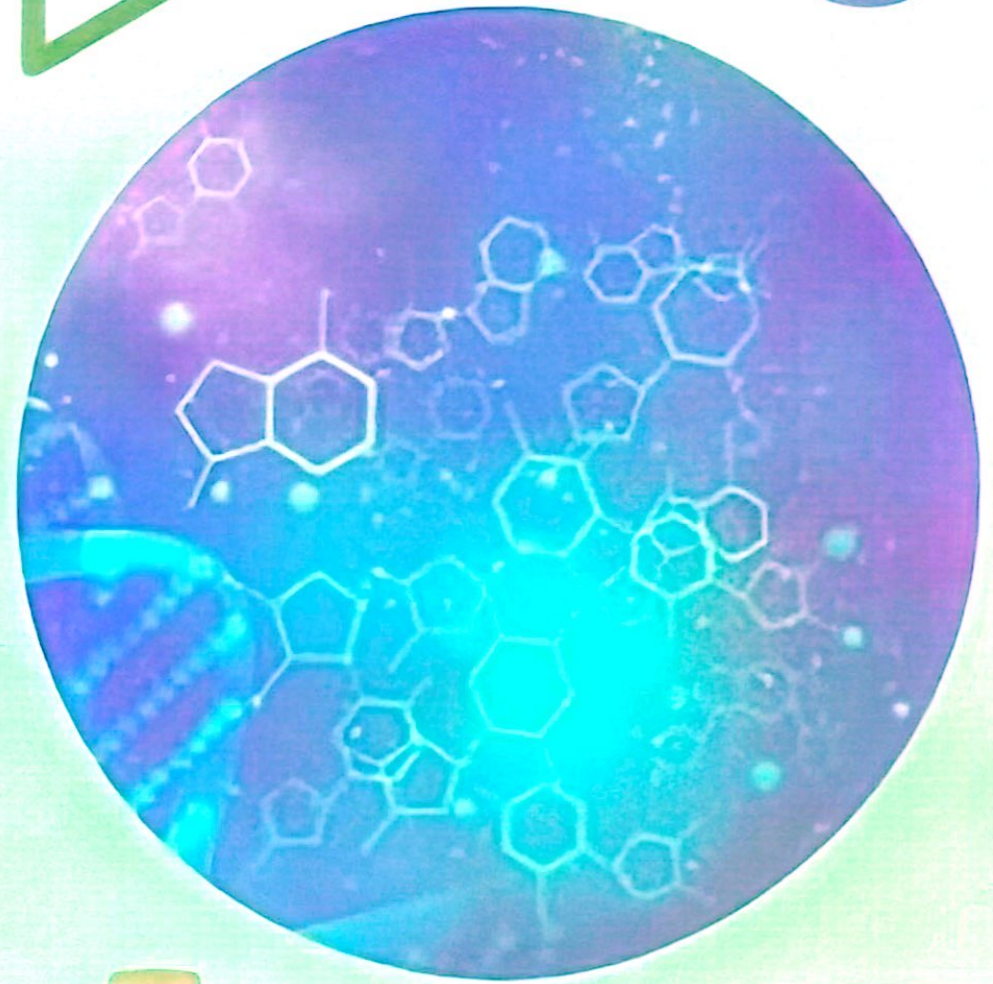
XX, X Y → without gonad
Gonadal dysgenesis

Phenotypic Intersex

Individuals have normal sex chromosomal and gonadal but abnormal reproductive tract and sexual characteristics

Molecular Genetics

Genetic material



The Genetic Material



Gene

- Link between generation
- Contain **genetic information**
- Pass to next generation and influence the form and characteristic of offspring
- Direct many complex processes leading to adult form and function

➤ Until 1944, not clear what chemical component of the cell constitute **genetic material** which contain **genetic information**

Characteristics of the Genetic Material



1. **Storage** of genetic Information
2. **Replication** of the genetic material
3. **Expression** of genetic Information
4. **Variation** by mutation

1. Storage of Genetic Information

The ability of genetic material to store genetic information is not necessarily directly related with its expression